Holmes large-L retention: completed frozen design and qualifications

No continuous commitment by 100,000 is an observed finite-horizon finding.

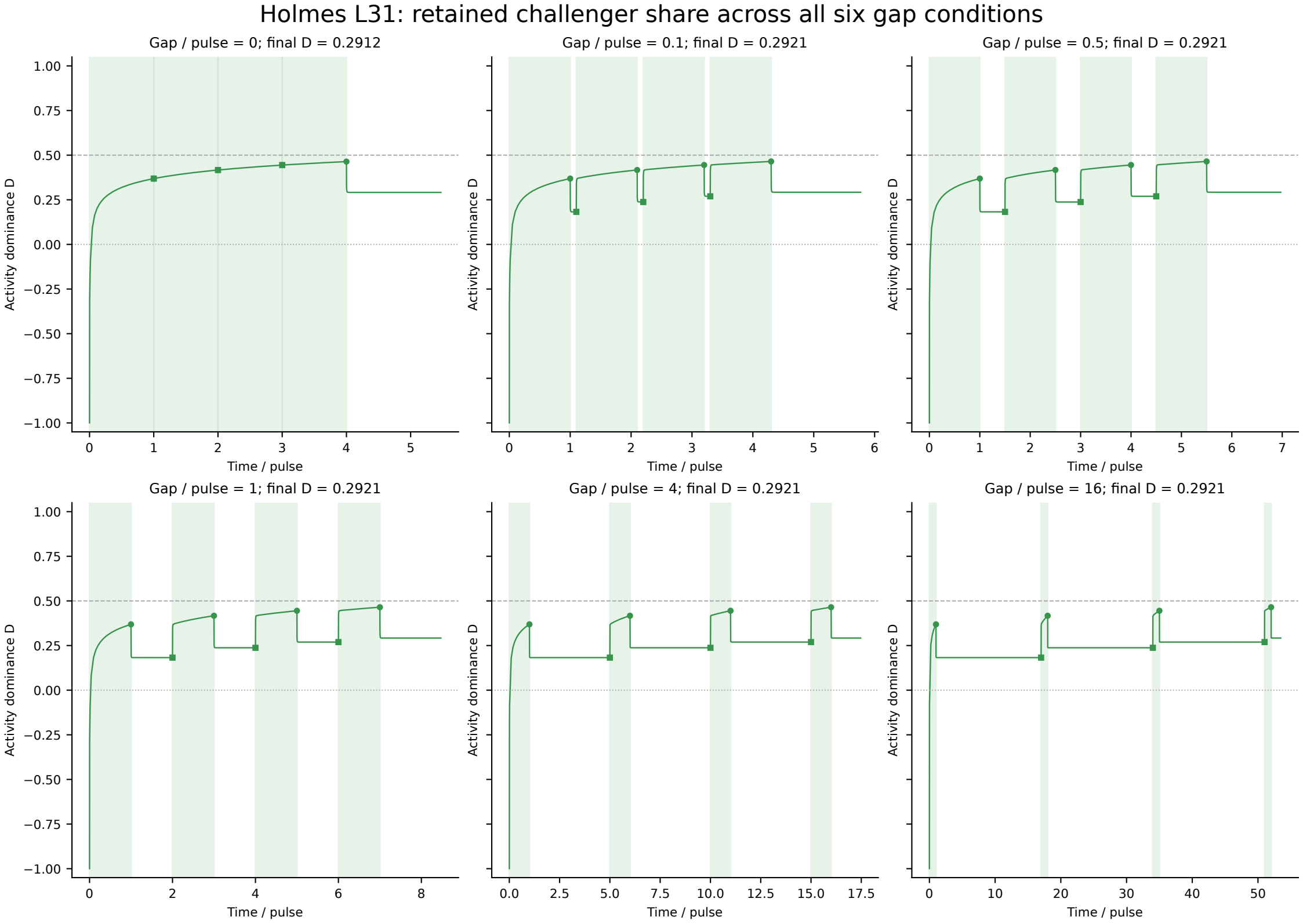
L27 upper committing duration = 18,125; four pulses each 6,796.875.
Target lengths: 31, 35, 39; gaps: 0, 0.1, 0.5, 1, 4, 16 × pulse.
Primary: D and challenger share at every pulse and gap endpoint. Fate is secondary.

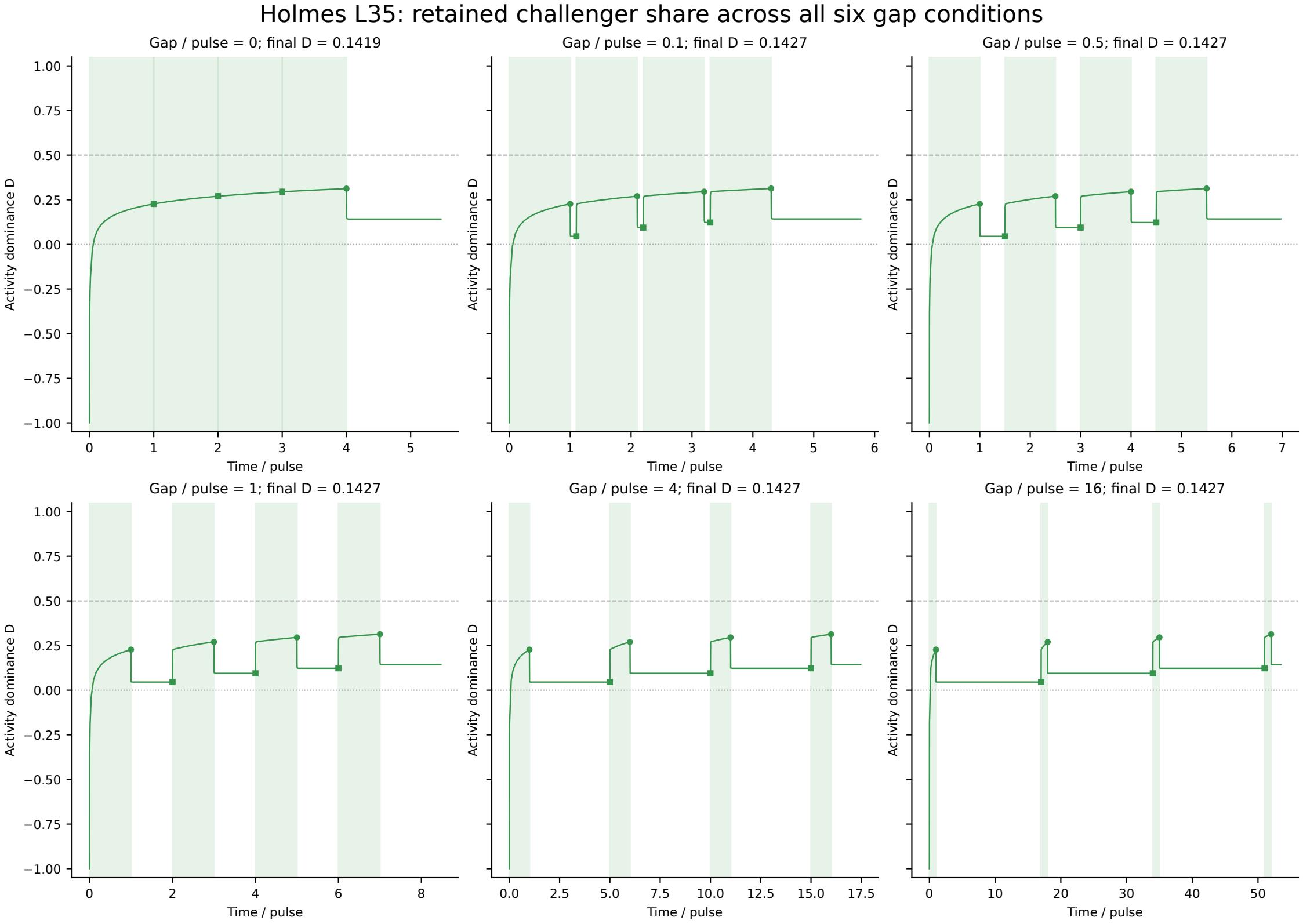
L39 bounded-noise check: gap16, mean 2% native peak, maximum 4%;
12 independent seeds plus matched-mean and zero controls.
S3-matched nonperiodic spatial OU input: ell 1.9, tau 10, knot 2.5.

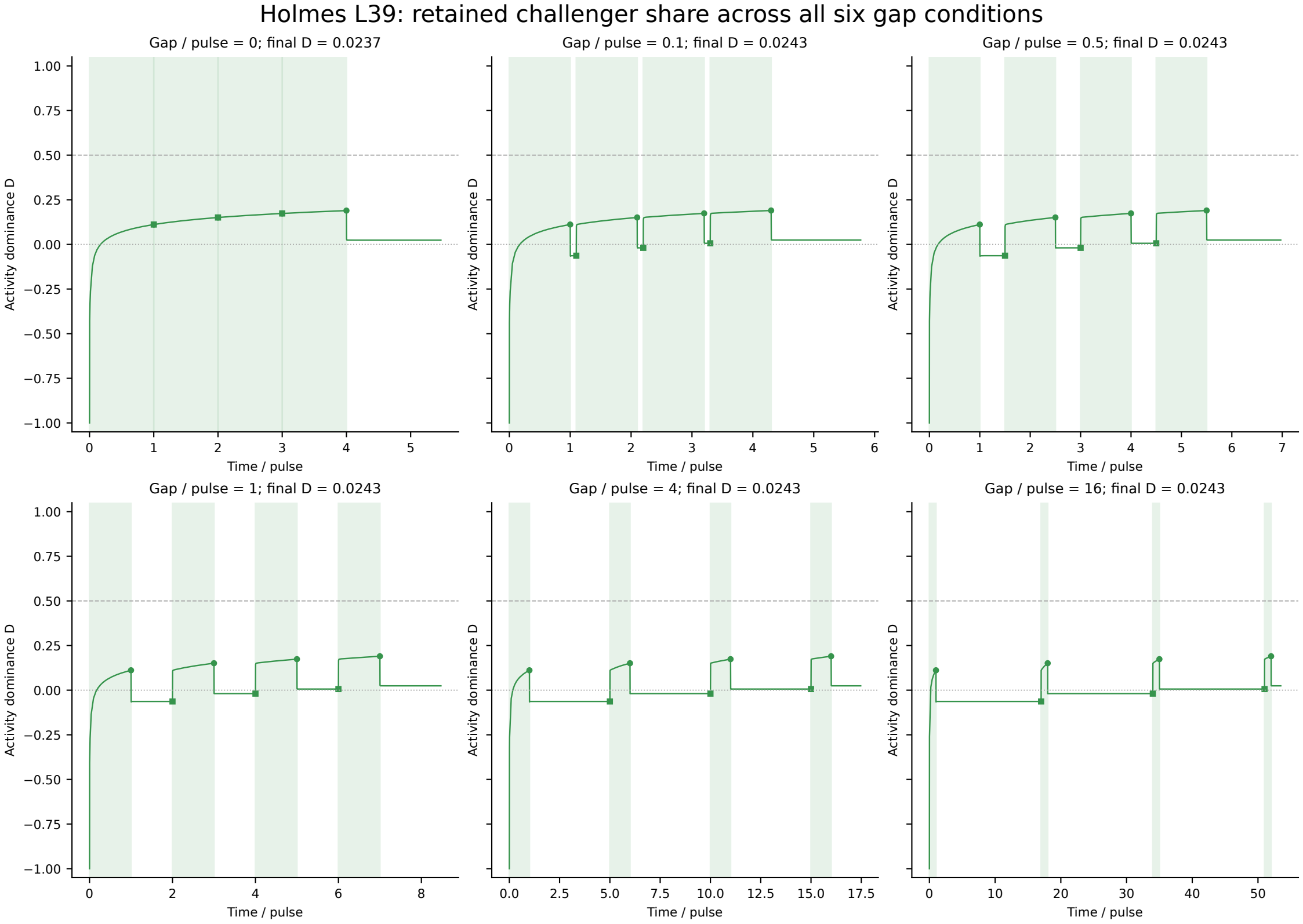
All 32 native SLURM trajectories and independent numeric/phase/restart/tail
audits completed. No kinetics or calibration thresholds were retuned.
Clean L39 gap4 retains >84% raw share even with slightly negative gap-end D.
The secondary S3 criterion requires both D ≥ 0 and raw retention ≥80%.

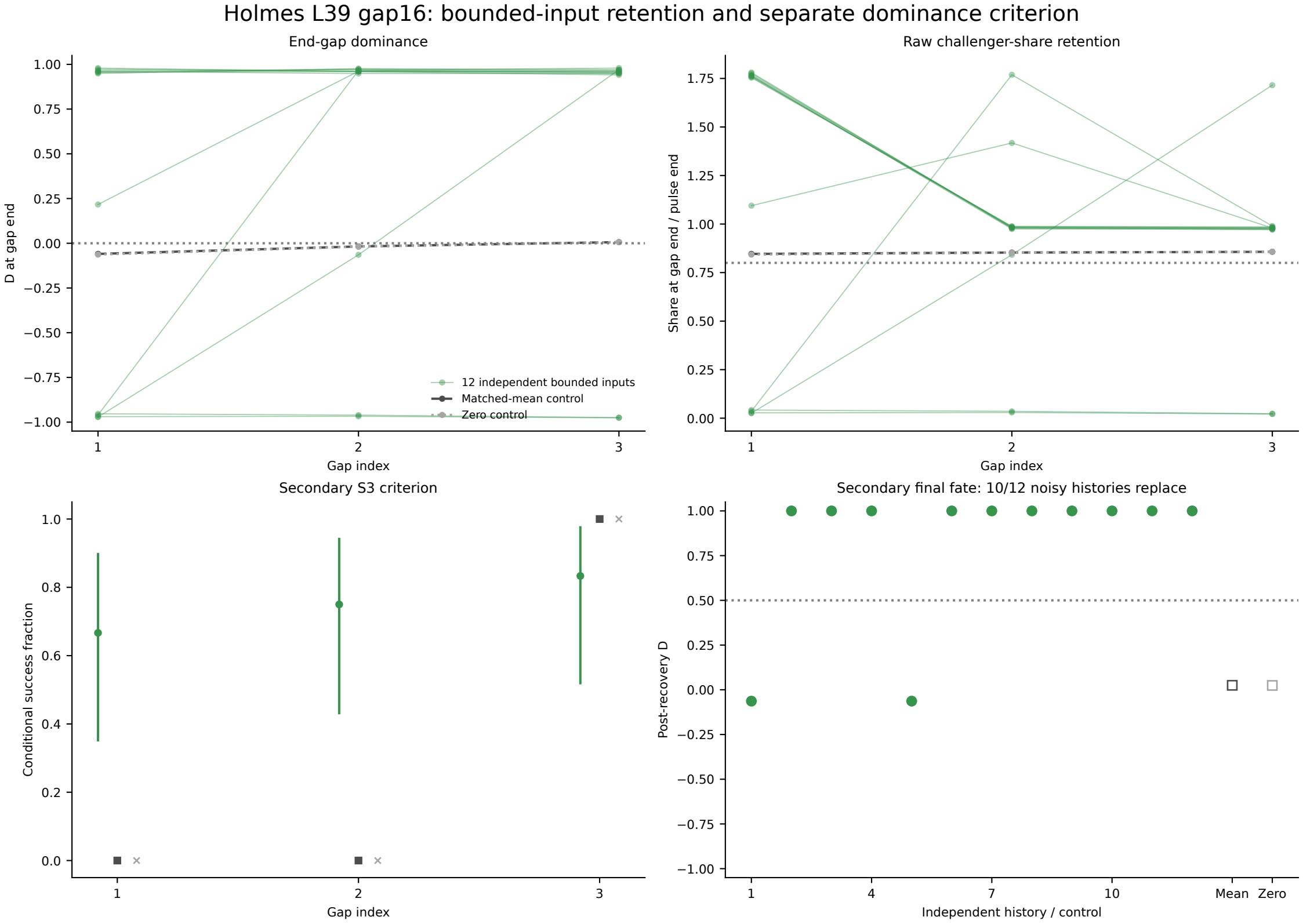
Class membership is unchanged. deBelly initiation remains interval-unresolved;
Holmes width-penalty growth is established only over measured moderate sizes.

The theoretical mean gap-integrated commanded input is 2.46 × ON exposure;
this long-gap assay is weak instantaneously, not a matched-total-dose test.
Heterogeneous input yields variable gap retention and 10/12 final replacements;
the matched-mean and zero controls remain below replacement.


Holmes L31: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.2912
Gap / pulse = 0.1; final D = 0.2921
Gap / pulse = 0.5; final D = 0.2921
Gap / pulse = 1; final D = 0.2921
Gap / pulse = 4; final D = 0.2921
Gap / pulse = 16; final D = 0.2921
Activity dominance D
Time / pulse


Holmes L35: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.1419
Gap / pulse = 0.1; final D = 0.1427
Gap / pulse = 0.5; final D = 0.1427
Gap / pulse = 1; final D = 0.1427
Gap / pulse = 4; final D = 0.1427
Gap / pulse = 16; final D = 0.1427
Activity dominance D
Time / pulse


Holmes L39: retained challenger share across all six gap conditions
Gap / pulse = 0; final D = 0.0237
Gap / pulse = 0.1; final D = 0.0243
Gap / pulse = 0.5; final D = 0.0243
Gap / pulse = 1; final D = 0.0243
Gap / pulse = 4; final D = 0.0243
Gap / pulse = 16; final D = 0.0243
Activity dominance D
Time / pulse


Holmes L39 gap16: bounded-input retention and separate dominance criterion
End-gap dominance
D at gap end
Gap index
12 independent bounded inputs
Matched-mean control
Zero control
Raw challenger-share retention
Share at gap end / pulse end
Gap index
Secondary S3 criterion
Conditional success fraction
Gap index
Secondary final fate: 10/12 noisy histories replace
Post-recovery D
Independent history / control
Mean
Zero

Holmes L31: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.3686983	0.3686983	0.4167275	0.4167275	0.4443979	0.4443979	0.4639236	0.2912072
0.1	0.3686983	0.1823237	0.4169269	0.2374473	0.4448783	0.2694141	0.4647032	0.2920993
0.5	0.3686983	0.1823238	0.4169269	0.2374478	0.4448786	0.2694155	0.4647042	0.2921004
1	0.3686983	0.1823239	0.4169270	0.2374485	0.4448790	0.2694172	0.4647053	0.2921017
4	0.3686983	0.1823247	0.4169273	0.2374523	0.4448812	0.2694278	0.4647122	0.2921096
16	0.3686983	0.1823278	0.4169287	0.2374675	0.4448900	0.2694698	0.4647399	0.2921412

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Holmes L35: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.2268125	0.2268125	0.2701031	0.2701031	0.2949934	0.2949934	0.3125254	0.1419210
0.1	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
0.5	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
1	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426672
4	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426673
16	0.2268125	0.0453329	0.2702584	0.0942794	0.2953922	0.1226072	0.3131874	0.1426673

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Holmes L39: all pulse and gap endpoint D values

Gap / pulse	Pulse 1	Gap 1	Pulse 2	Gap 2	Pulse 3	Gap 3	Pulse 4	Recovery
0	0.1113753	0.1113753	0.1508245	0.1508245	0.1734726	0.1734726	0.1894153	0.0236737
0.1	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
0.5	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
1	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
4	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877
16	0.1113753	-0.0632565	0.1509389	-0.0191915	0.1737945	0.0062700	0.1899665	0.0242877

Gap0 has three exact zero-duration inter-pulse boundaries. Full raw-share values, eligibility, S3 success, field tails and final fate are in holmes_large_L_endpoints.csv. Values are observed endpoints; the target-length commitment calibration is unavailable.

Class-membership evidence: Goryachev

initiation slows with L
Value: 2.898986378675809; bounds: [2.871794871794872, 2.9220779220779223]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: 1.124834416847597; bounds: [1.124484748557296, 1.1262376237623761]
L19 {'status': 'bounded', 'lower': 1.1700680272108843, 'upper': 1.1849315068493151, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.1798811734705885, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 1.124484748557296, 'upper': 1.1262376237623761, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.124834416847597, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; modest non-growing penalty is distinct from the large saturating-model penalty

share held through gaps
Value: 2.4681882341887874e-21; bounds: [None, None]
L39 gap4 first-gap D -0.58178128 to -1; share 0.20910936 to 5.1612126e-22; all raw retention ratios [2.4681882341887874e-21, 2.468188239315979e-21, 2.4681882814510725e-21]

commits after in-gap collapse
Value: -1.0; bounds: [None, None]
L39 gap4 gap-end D [-1.0, -1.0, -1.0]

Class-membership evidence: Wave-pinning

initiation slows with L
Value: 1.001354777811466; bounds: [1.0, 1.0070175438596491]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: None; bounds: [11.059907834101383, None]
L19 {'status': 'bounded', 'lower': 3.9870689655172415, 'upper': 4.049676025917927, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 4.023952792808644, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'lower_bound_only', 'lower': 11.059907834101383, 'upper': None, 'lower_strict': True, 'upper_strict': False, 'estimate_from_interpolation': None, 'narrow_censored': True, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}

share held through gaps
Value: 0.9656458108363115; bounds: [None, None]
L39 gap4 first-gap D 0.38625115 to 0.33862762; share 0.69312558 to 0.66931381; all raw retention ratios [0.9656458108363115, 0.9783668083224305, 1.2867515519314443]

commits after in-gap collapse
Value: 0.9999999753672253; bounds: [None, None]
L39 gap4 gap-end D [0.3386276200076397, 0.444706256945651, 0.9999999753670956]

Class-membership evidence: Holmes

initiation slows with L
Value: 1.001466041861598; bounds: [0.9848484848484849, 1.0153846153846153]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: None; bounds: [133.15579227696406, None]
Deciding comparison is L23 versus L19; extreme L39 remains uninformative. L19 {'status': 'bounded', 'lower': 28.99002493765586, 'upper': 29.125, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 29.034710721675975, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'lower_bound_only', 'lower': 0.0, 'upper': None, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': None, 'narrow_censored': True, 'broad_censored': True, 'informative': False, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; Holmes growth verified L19 ~29.03 to L23 >133.16; growth through L39 unresolved, not an assigned infinite ratio

share held through gaps
Value: 0.842868741871212; bounds: [None, None]
L39 gap4 first-gap D 0.11137532 to -0.063256484; share 0.55568766 to 0.46837176; all raw retention ratios [0.842868741871212, 0.8521812245120128, 0.8572795615775773]; first two gap D<0 fail the secondary S3 D>=0 criterion but retain >84% raw share

commits after in-gap collapse
Value: 0.024287736083136613; bounds: [None, None]
L39 gap4 gap-end D [-0.06325648449328491, -0.019191515710938245, 0.006269998360605156]

Class-membership evidence: deBelly

initiation slows with L
Value: None; bounds: [0.0, None]
deBelly first-update intervals do not resolve a normalized size effect

narrow-cue penalty and growth
Value: 0.963187372120211; bounds: [0.9548872180451128, 0.9696969696969697]
L19 {'status': 'bounded', 'lower': 0.9615384615384616, 'upper': 1.04, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.011603783514335, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 0.9548872180451128, 'upper': 0.9696969696969697, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 0.963187372120211, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}

share held through gaps
Value: 0.03626279229899842; bounds: [None, None]
L39 gap4 first-gap D 0.64353609 to -0.94040079; share 0.82176804 to 0.029799604; all raw retention ratios [0.03626279229899842, 0.038068019303912876, 0.23249058800910286]

commits after in-gap collapse
Value: 0.9164591847342345; bounds: [None, None]
L39 gap4 gap-end D [-0.9404007923089395, -0.9347088176848134, -0.5855576180817079]; deBelly first two collapse and final replacement agree with quarter-step final D 0.9208534696017405; later-gap timing, third-gap bias sign and endpoint fields remain sensitive

Class-membership evidence: Spring

initiation slows with L
Value: 0.9991692546590409; bounds: [0.8333333333333334, 1.2]
L39/L19, central area, fixed 10% L19 reporter

narrow-cue penalty and growth
Value: 1.1610062875918385; bounds: [1.1511627906976745, 1.1764705882352942]
L19 {'status': 'bounded', 'lower': 1.2, 'upper': 1.3571428571428572, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.2728575892292848, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; L39 {'status': 'bounded', 'lower': 1.1511627906976745, 'upper': 1.1764705882352942, 'lower_strict': False, 'upper_strict': False, 'estimate_from_interpolation': 1.1610062875918385, 'narrow_censored': False, 'broad_censored': False, 'informative': True, 'interpretation': 'Sampled-event/censoring bounds; no finite event time assigned to censoring.'}; modest non-growing penalty is distinct from the large saturating-model penalty

share held through gaps
Value: 0.003315145224882711; bounds: [None, None]
L39 gap4 first-gap D -0.26175354 to -0.99755261; share 0.36912323 to 0.0012236971; all raw retention ratios [0.003315145224882711, 0.021681739713766957, 1.0364261376049886]

commits after in-gap collapse
Value: 0.9996954239534162; bounds: [None, None]
L39 gap4 gap-end D [-0.9975526057568276, -0.976390262546511, 0.9966415640033679]